Auctions
Beyond single-item

COMP6203 Intelligent Agents

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Learning Outcomes 1

By the end of this lecture, you should be able to

- **Define** multi-unit auctions and combinatorial auctions settings
- **Describe** three different auctions for multi-unit single-unit demand setting
- **Compute** the outcome (allocation and payment) or running any of these three multi-unit single-unit demand auctions
- **Describe** the Revelation Equivalence theorem for multi-unit case and explain its implications

Previously

- Learned about single-item auctions
  - One item is put up for sale
  - Information from this sale is not carried forward to future auctions
- Made some simplifying assumptions
  - Independent private value model
  - Bidders cannot collude
- Set the objective to be maximising the social welfare
  - Award to the bidder with the highest value

Recall: Sealed-bid single-item auctions

- First-price auction
  - Shade your bid: put a bid lower than your valuation.
  - How much lower? Do the math!

- Second-price (Vickrey) auction
  - Placing a bid equal to your valuation is a dominant strategy → Vickrey auction is (dominant strategy) truthful.

Multi-unit auctions with single-unit demand

- There are *k* identical items for sale in a single auction (i.e., one auction)
- There are *n* strategic risk-neutral bidders
- Each bidder only wants one unit
- Bidders have independent private values for these single units

- What auction should we run?
  - Which bidders should each be allocated one of the items
  - How much should each of the winners pay

Your thoughts

Let’s start with 2 items

- There are 2 identical items (say first edition of a book) for sale in a single auction
- There are *n* strategic bidders
- Every bidder only wants one unit

- What auction should we run?
  - Which two bidders should each be allocated one of the items
  - How much should each of the winners pay

VCG: Vickrey extended to beyond single-item

Assume there are *k* identical items for sale, each bidder wants one unit

- The top *k* bidders each win one item
- The payment of each winning bidder is equal to the k+1st highest bid

- Is VCG efficient? I.e., does it maximise social welfare?
- Is VCG (dominant strategy) truthful?

Example

What is the revenue of VCG when:

- $k = 1$
  - £20
- $k = 2$
  - £30
- $k = 3$
  - £24
- $k = 4$
  - £0

Bidder	Bid amount
1	£25
2	£20
3	£15
4	£8

Some other auctions

- Pay-your-bid: Discriminatory pricing, because bidders will pay different amounts for the same thing.
  - ❖ Generalises first-price auction
- Lowest winning bid: Similar to VCG but ensures that bidders don't pay zero if there are fewer bids than units for sale.
  - Used in eBay
  - Is it truthful?
  - How else can we ensure bidders don't pay zero if there are fewer bids than units for sale, but ensure truthfulness?
- Sequential auctions: repeated single-item auctions Next Lecture

Recall: Revenue Equivalence

Assume that all bidders are risk-neutral and utility maximisers, and each has an independent private valuation for the single item, drawn from a common cumulative distribution $F(\theta)$ that is strictly increasing and atomless on $[L,H]$. Then any auction mechanism in which

- in equilibrium, the item will be allocated to the agent with the highest valuation, and
- any agent with valuation L has an expected utility of zero;

yields the same expected revenue, and hence results in any bidder with valuation θ making the same expected payment.

First-price, second-price, English and Dutch auctions all satisfy the above. Hence, they are **revenue equivalent**.

Revenue Equivalence for multi-unit auctions

Assume that all bidders are risk-neutral and utility maximisers, and each has an independent private valuation for the single item, drawn from a common cumulative distribution $F(\theta)$ that is strictly increasing and atomless on $[L,H]$. Then any auction mechanism in which

- in equilibrium, the items are allocated efficiently (i.e. the k bidders with the highest valuation each win one item), and
- any agent with valuation L has an expected utility of zero;

yields the same expected revenue, and hence results in any bidder with valuation θ making the same expected payment.

All payment rules discussed earlier (VCG, pay-your-bid, lowest winning bid) satisfy the above. Hence, they are **revenue equivalent**.

- If assumptions of the theorem hold: bidders are risk-neutral and play equilibrium strategies

(general) Multi-unit auctions

- There are *k* identical items for sale (e.g. general admission ticket to a concert) in a single auction (i.e., one auction)
- There are *n* strategic risk-neutral bidders
- Each bidder wants one or more units
- Bidders have independent private values for these single units

- What auction should we run?
  - Which bidders should each be allocated one of the items
  - How much should each of the winners pay

Not examinable: computing payments in this truthful mechanism for general multi-unit auctions

Example: concert tickets

- Five tickets for sale, 4 bidders interested in purchasing 1 or 2 tickets
- What allocation is efficient (social welfare maximizing)?
  - Bidders 1 and 2 each get 2 tickets, bidder 3 gets 1 ticket

Bidder	Valuation for 1 ticket	Valuation for 2 tickets
1	£25	£50
2	£20	£40
3	£15	£30
4	£8	£16

- What should each winner pay for the mechanism to be dominant strategy truthful?
  - Bidder 1 pays £(20*2+15*2+8)-£(20*2+15) = 78-55= £23
  - Bidder 2 pays £(25*2+15*2+8)-£(25*2+15) = 88-65= £23
  - Bidder 3 pays £(25*2+20*2+8)-£(25*2+20*2) = 98-90= £8

Bidder 1 and 2 pay £11.5 per ticket, Bidder 3 pays £8 for a ticket

Other payment functions: DS or not?

- Five tickets for sale, 4 bidders interested in purchasing 1 or 2 tickets
- Efficient allocation: Bidders 1 and 2 each get 2 tickets, bidder 3 gets 1 ticket
- The following payments do not incentivise agents to bid truthfully. Why?
  - Every ticket is charged £8
  - Bidder 1 pays £40, bidder 2 pays £30, bidder 3 pays £8
  - Bidder 1 and 2 both pay £30 each, bidder 3 pays £8

Bidder	Valuation for 1 ticket	Valuation for 2 tickets
1	£25	£50
2	£20	£40
3	£15	£30
4	£8	£16

Think about each agent's utility.
Identify at least one agent that can benefit from mireporting.

VCG for general multi-unit auctions

Assume there are *k* identical items for sale, each bidder wants one or more units

- No longer k+1st-price auction
- All winning bidders who won the same number of units with pay the same amount as each other
- Bidders who win different numbers of units will not necessarily pay the same per unit prices
- Bidders who win large number of units will pay at least as much in total (not necessarily per unit) as bidders who won smaller numbers of units

Revisit Example: concert tickets

- Five tickets for sale, 4 bidders interested in purchasing 1 or 2 tickets
- Efficient allocation gives bidder 1 and 2 each two tickets, and bidder 3 one ticket
- Why charging each winner the highest losing bid (here £15) per awarded ticket is not truthful?
  (note that 3 is happy to purchase 2 tickets so highest losing bid is £15)
- Bidding truthfully:
  - Bidder 1's utility: £(25*2-15*2)=50-30=£20
  - Bidder 2's utility: £(20*2-15*2)=40-30=£10
  - Bidder 3's utility: £(15*1-15*1)=£0
- What if bidder 3 places bid equal to £9 (and the others bid truthfully)?
  - The efficient allocation still awards bidder 3 one ticket, but charges £9 for it. Therefore bidder 3's utility is increased to £(15*1-9*1)=£6.
- Can bidder 1 or 2 benefit from misreporting (while others bid truthfully)?
  - Not bidder 1 (not in this example): no change until bidding below £15. Bidding between £8 and £15 results in one ticket for bidder 1 at the price of their bid (so utility at most £25-£8=£13). Bidding below £8 yields utility zero for bidder 1.
  - Bidder 2 can benefit by placing a bid just a little above £8 (say £9). The efficient allocation will award him one ticket at the price of £9, yielding bidder 2 utility £20-£9=£11 which is higher than £10 obtained from truthful bidding.

Bidder	Valuation for 1 ticket	Valuation for 2 tickets
1	£25	£50
2	£20	£40
3	£15	£30
4	£8	£16